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Liquid collector

Abstract

An embodiment of the present disclosure provides a liquid collector that collects liquid, the liquid collector including: a flow passage member including a flow passage that connects one end and other end of the flow passage member; a contact member that is disposed at the one end of the flow passage member, that includes an elastic material, and that is configured to come into contact with a target from which liquid is to be collected; and a container that receives the liquid discharged from the flow passage at the other end of the flow passage member.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is the National Stage entry of International Application No. PCT/JP2021/020577, filed 30 May 2021, which claims priority to Japanese Application No. 2020-095795, filed 1 Jun. 2020.

BACKGROUND

Technical Field

(2) The present disclosure relates to a liquid collector.

Description of Related Art

(3) A small amount of liquid may be collected by, for example, suction using a dropper or a syringe or absorption into test paper.

(4) It is desirable to collect liquid efficiently. However, when, for example, a small amount of liquid is collected by using a dropper or the like, it may be difficult to efficiently collect all of the liquid.

Summary of Invention

(5) Here, it is recognized, for example, without limitation, that efficient collection of liquid is desired.

SUMMARY

(6) Some embodiments of the present disclosure provide a liquid collector. In some embodiments, the liquid collector collects liquid. In some embodiments, the liquid collector includes a flow passage member. In some embodiments, the flow passage member includes a flow passage that connects one end and other end of the flow passage member. In some embodiments, the liquid collector includes a contact member. In some embodiments, the contact member is disposed at the one end of the flow passage member. In some embodiments, the contact member includes an elastic material. In some embodiments, the contact member is configured to come into contact with a target from which liquid is to be collected. In some embodiments, the liquid collector includes a container. In some embodiments, the container receives the liquid discharged from the flow passage at the other end of the flow passage member.

(7) Some embodiments of the present disclosure provide a liquid collector. In some embodiments, the liquid collector collects liquid. In some embodiments, the liquid collector includes a flow passage member. In some embodiments, the flow passage member includes a plurality of flow passages that connect one end and other end of the flow passage member. In some embodiments, the liquid collector includes a container. In some embodiments, the container receives the liquid discharged from the flow passages at the other end of the flow passage member.

(8) According to the above-described embodiments, for example, the liquid can be efficiently collected.

(9) Additional aspects and advantages of the present disclosure will become readily apparent to those skilled in this art from the following detailed description, wherein only illustrative embodiments of the present disclosure are shown and described. As will be realized, the present disclosure is capable of other and different embodiments, and its several details are capable of modifications in various obvious respects, all without departing from the disclosure. Accordingly, the drawings and description are to be regarded as illustrative in nature, and not as restrictive.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a sectional view of a liquid collector according to an embodiment.

- (2) FIG. 2 is a sectional view of a flow passage member of the liquid collector illustrated in FIG. 1.
- (3) FIG. 3 illustrates the manner in which liquid is collected in the liquid collector illustrated in FIG. 1.
- (4) FIG. 4 is a sectional view of a liquid collector according to an embodiment.
- (5) FIG. 5 illustrates the manner in which liquid is collected in the liquid collector illustrated in FIG. 4.
- (6) FIG. 6 is a sectional view of a liquid collector according to an embodiment.
- (7) FIG. 7 illustrates the manner in which liquid is collected in the liquid collector illustrated in FIG. 6.
- (8) FIG. 8 is a sectional view of a liquid collector according to an embodiment.
- (9) FIG. 9 illustrates the manner in which liquid is collected in the liquid collector illustrated in FIG. 8.
- (10) FIG. 10 is a sectional view of a liquid collector according to an embodiment.
- (11) FIG. 11 illustrates the manner in which liquid is collected in the liquid collector illustrated in FIG. 10.
- (12) FIG. 12 is a sectional view of a flow passage member of a liquid collector according to an embodiment.
- (13) FIG. 13 is a sectional view of a flow passage member of a liquid collector according to an embodiment.
- (14) FIG. 14 is a sectional view of a flow passage member of a liquid collector according to an embodiment.
- (15) FIG. 15 is a sectional view of a flow passage member of a liquid collector according to an embodiment.
- (16) FIG. 16 is a sectional view of a flow passage member of a liquid collector according to an embodiment.

DESCRIPTION OF EMBODIMENTS

- (17) In some embodiments, a liquid collector collects liquid (sometimes also referred to as solution).
- (18) In some embodiments, the liquid may be body fluid produced by secretion in a target subject, or liquid other than body fluid. The liquid other than body fluid may be liquid adhering to a target object or liquid not adhering to a target object. The liquid not adhering to a target object may be liquid contained in a target object.
- (19) The liquid to be collected may be a solution. The liquid may be body fluid, a solution derived from body fluid, or diluted body fluid. The liquid may be a solution other than (not derived from) body fluid, or a mixture of body fluid or a solution derived from body fluid with a solution not derived from body fluid. The solution may be a solution used for sample measurement or a solution used for calibration measurement. For example, the solution may be a standard solution or a calibration solution. The sample to be measured may be a specimen.
- (20) Examples of the body fluid include lymph fluid, tissue fluid such as inter-tissue fluid, intercellular fluid, and interstitial fluid, coelomic fluid, serous cavity fluid, pleural fluid, ascites fluid, pericardial fluid, cerebrospinal fluid, synovial fluid, and aqueous humor. Examples of the body fluid further include digestive juice, such as saliva, gastric juice, bile, pancreatic juice, and intestinal juice, sweat, tear, nasal mucus, urine, semen, vaginal fluid, amniotic fluid, and milk. The body fluid may be body fluid of an animal or a human. The “body fluid” may be a solution. The solution may include a physiological buffer, such as phosphate-buffered saline (PBS) or N-tris(hydroxymethyl)methyl-2-aminoethanesulfonic acid (TES) buffer, containing a measurement target substance. The solution is not particularly limited as long as the measurement target substance is contained.
- (21) In some embodiments, the target subject may include, or may be, a human. In some embodiments, the target subject may include, or may be, a non-human animal. The non-human

animal may include, or may be, a mammal. Examples of non-human animals include, but are not limited to, working animals, livestock animals, pet animals, and wild animals.

(22) In some embodiments, the liquid collector includes a flow passage member. In some embodiments, the flow passage member may include a flow passage that connects one end and other end of the flow passage member. In some embodiments, the flow passage member may include a plurality of flow passages or only one flow passage. In some embodiments, when the flow passage member includes a plurality of flow passages, the flow passages extend in a pipe axis direction connecting the one end and the other end of the flow passage member. In some embodiments, the flow passages may or may not be substantially parallel to each other.

(23) The term “pipe axis direction” used in this specification means a direction connecting the one end and the other end of the flow passage member.

(24) In some embodiments, the flow passage member may have a columnar shape that extends in the pipe axis direction connecting the one end and the other end of the flow passage member. In some embodiments, the flow passage member having the columnar shape may have a plurality of grooves or only one groove that extends in the pipe axis direction along an outer peripheral surface of the flow passage member. In some embodiments, the groove or grooves each serves as a flow passage for the liquid. The cross-sectional shape of the groove or grooves may be, for example, without limitation, a U-shape (the corners may be substantially right angled, obtuse angled, acute angled, or rounded), a V-shape (the corner may be substantially acute angled or rounded), or a general trough shape. In some embodiments, the groove or grooves may extend straight in the pipe axis direction. In some embodiments, the groove or grooves may not extend straight in the pipe axis direction. For example, the groove or grooves may be formed in the flow passage member in a helical shape. For example, the groove or grooves may be inclined with respect to the pipe axis direction at the one end at which the liquid is absorbed.

(25) In some embodiments, the flow passage member may include an outside tube. In some embodiments, the outside tube may have a tubular shape that extends in the pipe axis direction. In some embodiments, the flow passage member may include an inside member. In some embodiments, the inside member may be inserted in the outside tube. In some embodiments, the inside member may have a plurality of grooves that extend in the pipe axis direction along an outer peripheral surface of the inside member. In some embodiments, an inner peripheral surface of the outside tube and the grooves in the inside member may define the flow passages. In some embodiments, the outside tube may cover at least a portion of the inside member in the pipe axis direction. In some embodiments, the outside tube may cover the entirety of the inside member in the pipe axis direction. In some embodiments, the outside tube may cover a portion of the inside member in the pipe axis direction.

(26) In some embodiments, the flow passage member may have a columnar shape that extends in the pipe axis direction. In some embodiments, the flow passage member having the columnar shape may have a circular shape or an elliptical shape in cross-section when viewed in the pipe axis direction. In some embodiments, the flow passage member may have, for example, a polygonal shape, such as a triangular shape, a quadrangular shape, a pentagonal shape, a hexagonal shape, or an octagonal shape in cross-section. In some embodiments, the flow passage member may have a plurality of flow passages extending therethrough in the pipe axis direction to provide communication between the one end and the other end of the flow passage member. In some embodiments, the flow passage member may have a uniform cross-sectional shape in the pipe axis direction (for example, a cylindrical shape) or a cross-sectional shape that varies in the pipe axis direction (for example, decreases in size toward the one end).

(27) In some embodiments, the flow passage member may be formed by bundling a plurality of single pipes that extend in the pipe axis direction. In some embodiments, each single pipe may have a tubular shape that extends in the pipe axis direction and include a flow passage that connects one end and the other end of the single pipe. In some embodiments, the single pipes may be joined

together by adhesion or fusion bonding. In some embodiments, the single pipes may be bundled together with a band or the like. In some embodiments, the single pipes may be formed integrally with each other by using a resin material. When, for example, the single pipes are bundled together, holes in the single pipes and gaps between the single pipes serve as flow passages for the liquid.

(28) In some embodiments, the flow passage member may include a porous material. The porous material may be, for example, without limitation, a porous ceramic or a sponge. The porous material may serve as a flow passage that allows fluid to flow therethrough by capillarity.

(29) In some embodiments, the flow passage member may include an outer tube. In some embodiments, the outer tube may have a tubular shape that extends in the pipe axis direction. In some embodiments, the flow passage member may include a plurality of inner pipes. In some embodiments, the inner pipes may be inserted in the outer tube. In some embodiments, the inner pipes may extend in the pipe axis direction. When, for example, the inner pipes are disposed in the outer tube, holes formed in the inner pipes, gaps between the inner pipes, and gaps between an inner peripheral surface of the outer tube and the inner pipes disposed inside the inner peripheral surface serve as flow passages for fluid.

(30) In some embodiments, the liquid collector includes a contact member. In some embodiments, the contact member may be disposed at the one end of the flow passage member. In some embodiments, the contact member may be configured to come into contact with a target from which the liquid is to be collected. In some embodiments, the contact member may include, or may be, an elastic material. In some embodiments, when the contact member includes an elastic material, the contact member may be a combination of a portion made of the elastic material and a portion made of another material. In some embodiments, the contact member may have mechanical, physical, chemical, and biological characteristics, such as elasticity and surface shape, such that the contact member can be brought into contact with, for example, a skin, a mucous membrane, or an eyeball of an animal, such as a human, substantially without causing damage thereto. In some embodiments, the elastic material that forms the contact member may be softer than the flow passage member.

(31) The term “elastic material” used in this specification generally refers to a material that is elastically deformable when the contact member is pressed against a target subject or a target object. The elastic material can be brought into contact with a target subject or a target object substantially without causing damage to a surface thereof in normal use. In some embodiments, the elastic material may include a material such as a silicone-based material, an urethane-based material (foam type, low-rebound material), a rubber material, a synthetic resin material (e.g., polyethylene), or a combination thereof (for example, without limitation, a structure including an inner portion made of low-rebound urethane and an outer portion made of silicone), or be formed mainly of a material that is one of, or a combination of, the above-mentioned materials.

(32) In some embodiments, the contact member may include, or be made of, a hydrophilic material. In some embodiments, a contact surface may be hydrophilic. The contact member may have a hydrophilized surface. In some embodiments, the contact surface may be hydrophobic or water repellent. In some embodiments, the contact surface may be rough. In such a case, for example, a user (target subject) may recognize that the contact surface is in contact with their skin. In some embodiments, the contact surface may be embossed.

(33) In some embodiments, the contact member may include an inlet opening. In some embodiments, the inlet opening may come into contact with a surface of the target from which the liquid is to be collected and allow introduction of the liquid that is on or near the surface. In some embodiments, the contact member may include an outlet opening. In some embodiments, the outlet opening may be connected to the inlet opening to provide fluid communication. In some embodiments, the outlet opening may be connected to the flow passage member to provide fluid communication. In some embodiments, the one end of the flow passage member may be fitted to the outlet opening in the contact member. In some embodiments, the contact member may include a

sleeve. In some embodiments, the sleeve may have a tubular shape that extends toward the other end of the flow passage member. When, for example, the contact member includes the sleeve having a tubular shape that extends toward the other end of the flow passage member, the flow passage member is fitted to and covered with the sleeve. In some embodiments, the inlet opening and the outlet opening may have the same size or shape. In some embodiments, the inlet opening and the outlet opening may have different sizes or shapes. For example, the inlet opening may be relatively wide, and the thickness of a flow passage that provides fluid communication between the inlet opening and the outlet opening may be reduced toward the outlet opening. This arrangement may be reversed.

(34) In some embodiments, the contact member may have a contact surface in a region around the inlet opening. In some embodiments, the contact surface may come into contact with a surface of the target from which the liquid is to be collected. In some embodiments, the contact surface may protrude from the one end of the flow passage member. In some embodiments, at least a portion of the contact surface may be spherical, or the entirety of the contact surface may be spherical.

(35) In some embodiments, the liquid collector includes a container. In some embodiments, the container may receive the liquid discharged from the flow passage or flow passages at the other end of the flow passage member. In some embodiments, the container may include a peripheral wall having a tubular shape. In some embodiments, the peripheral wall may extend in the pipe axis direction. In some embodiments, the container may include a bottom portion. In some embodiments, the bottom portion may block the peripheral wall at one side in the pipe axis direction. For example, the container may be formed of the peripheral wall and the bottom portion so as to have a tubular shape with a bottom. In some embodiments, the bottom portion of the container may be spaced from the other end of the flow passage member in the pipe axis direction. In some embodiments, a reservoir space that stores the liquid may be defined between the bottom portion of the container and the other end of the flow passage member. In some embodiments, the container may have a shape of a dish, a bowl, a tray, etc., having a recess capable of receiving the liquid.

(36) In some embodiments, the container may have an opening at a side opposite to the side at which the bottom portion is provided in the pipe axis direction. In some embodiments, the contact member may be disposed to close the opening of the container. In some embodiments, the contact member may be mechanically connected to the opening of the container to close the opening of the container. In some embodiments, the contact member may be placed on the opening of the container to close the opening of the container without being mechanically connected to the opening of the container.

(37) In some embodiments, the peripheral wall of the container may have an inner diameter that gradually decreases toward the bottom portion in the pipe axis direction. In some embodiments, the other end of the flow passage member may be in contact with an inner peripheral surface of the peripheral wall of the container. When, for example, the other end of the flow passage member is in contact with the inner peripheral surface of the peripheral wall, the liquid in the flow passage or flow passages of the flow passage member easily moves to the inner peripheral surface of the peripheral wall.

(38) In some embodiments, the container may or may not have an air hole that provides communication with the outside of the container. In some embodiments, the air hole may be a through hole connecting the inside and the outside of the container, or a gap between the container and the contact member. In the case where, for example, the container has the air hole, an amount of air corresponding to the volume of the liquid that flows into the container is discharged to the outside of the container through the air hole when the liquid flows into the container through the flow passage or flow passages. In the case where, for example, the container has no air hole, an amount of air corresponding to the volume of the liquid that flows into the container is discharged to the outside of the container through one or more of the flow passages when the liquid flows into

the container through the flow passages.

(39) In some embodiments, the air hole may be formed at a position such that the liquid that flows into the container does not flow out of the container when the liquid collector is used. In some embodiments, the air hole may be sealable. For example, a plug or seal that closes the air hole from the outside of the container may be provided. The air hole may be closed to seal the inside of the container. Accordingly, for example, without limitation, the liquid collected in the container can be prevented from leaking to the outside.

(40) In some embodiments, the container may have a scale mark that indicates an amount of the liquid received in the container. When, for example, the container has the scale mark, the amount of the liquid received in the container can be easily visually recognized. In some embodiments, the scale mark may be disposed to enable estimation of the amount of the liquid received in the container. In some embodiments, the scale mark may be disposed at a position corresponding to a predetermined amount (required amount). Accordingly, for example, it can be determined whether a sufficient amount of liquid is collected. One, two, or more scale marks may be provided.

(41) In some embodiments, the container may include a cone-shaped portion. In some embodiments, the cone-shaped portion may be disposed in the container. In some embodiments, at least a portion of the cone-shaped portion in the pipe axis direction may have an inner diameter that gradually decreases toward the bottom portion in the pipe axis direction. In some embodiments, the cone-shaped portion may be formed over at least a partial region in the pipe axis direction, or over the entire region in the pipe axis direction. In some embodiments, the cone-shaped portion may be a component separate from the container or a part of the container. In some embodiments, when the cone-shaped portion is provided, the container may include no bottom portion and be open. In some embodiments, the other end of the flow passage member may be in contact with an inner peripheral surface of the cone-shaped portion. When, for example, when the other end of the flow passage member is in contact with the inner peripheral surface of the cone-shaped portion, the liquid in the flow passage or flow passages of the flow passage member easily moves to the inner peripheral surface of the cone-shaped portion. For example, when the other end of the flow passage member is in contact with the inner peripheral surface of the cone-shaped portion, movement of the flow passage member toward the bottom portion of the container is restricted, and the flow passage member is positioned with respect to the container in the pipe axis direction.

(42) In some embodiments, the liquid collector may include a flow passage member and a container. In some embodiments, the flow passage member may include a plurality of flow passages that connect one end and other end of the flow passage member. In some embodiments, the container may receive liquid discharged from the flow passages at the other end of the flow passage member.

(43) FIGS. **1** to **3** illustrate a liquid collector **100** according to an embodiment. As illustrated in FIGS. **1** to **3**, the liquid collector **100** includes a flow passage member **110**, a contact member **120**, and a container **130**. The liquid collector **100** collects liquid.

(44) The flow passage member **110** includes an outside tube **111** and an inside member **112**. The outside tube **111** has a tubular shape that extends in a pipe axis direction D_s connecting one end **1101** and other end **1102** of the flow passage member **110**. In the embodiment of the present disclosure, the outside tube **111** has, for example, a circular shape in cross-section when viewed in the pipe axis direction D_s . The outside tube **111** may also have, for example, an elliptical shape in cross-section when viewed in the pipe axis direction D_s . The outside tube **111** may also have, for example, a polygonal shape, such as a triangular shape, a quadrangular shape, a pentagonal shape, a hexagonal shape, or an octagonal shape in cross-section when viewed in the pipe axis direction D_s .

(45) The inside member **112** is inserted in the outside tube **111**. The inside member **112** has a columnar shape that extends in the pipe axis direction D_s . The inside member **112** has a uniform cross-sectional shape in the pipe axis direction D_s . The inside member **112** has a plurality of grooves **1121** that extend in the pipe axis direction D_s . The grooves **1121** are arranged with equal

intervals therebetween in a circumferential direction around a central axis of the inside member **112**. Each groove **1121** is recessed radially inward from an outer peripheral surface **1122** of the inside member **112**. Each groove **1121** extends continuously in the pipe axis direction Ds.

(46) The outer peripheral surface **1122** of the inside member **112** inserted in the outside tube **111** is in contact with, or slightly spaced from, an inner peripheral surface **1111** of the outside tube **111**. In the flow passage member **110**, flow passages **113** are defined between the inner peripheral surface **1111** of the outside tube **111** and the grooves **1121** in the inside member **112**. The flow passages **113** extend in the pipe axis direction Ds to connect the one end **1101** and the other end **1102** of the flow passage member **110**.

(47) Thus, the flow passage member **110** includes a plurality of flow passages **113**. The flow passages **113** extend in the pipe axis direction Ds connecting the one end **1101** and the other end **1102**. The flow passages **113** are substantially parallel to each other.

(48) The contact member **120** is disposed at the one end **1101** of the flow passage member **110**. The contact member **120** is configured to come into contact with a target from which the liquid is to be collected. The contact member **120** may include, or may be, an elastic material. When the contact member **120** includes an elastic material, the contact member **120** may be a combination of a portion made of the elastic material and a portion made of another material.

(49) The contact member **120** has a flow-passage-member receiving hole **121**. The flow-passage-member receiving hole **121** extends in the pipe axis direction Ds. The flow-passage-member receiving hole **121** has an inlet opening **1211** at an end thereof at one side in the pipe axis direction Ds. The inlet opening **1211** opens at a side opposite to a side at which the flow passage member **110** is disposed in the pipe axis direction Ds. The inlet opening **1211** comes into contact with a surface of the target from which the liquid is to be collected and allows introduction of the liquid that is on or near the surface.

(50) The flow-passage-member receiving hole **121** has an outlet opening **1212** at an end thereof at the other side in the pipe axis direction Ds. The outlet opening **1212** opens at the side at which the flow passage member **110** is disposed in the pipe axis direction Ds. The outlet opening **1212** is connected to the inlet opening **1211** to provide fluid communication through the flow-passage-member receiving hole **121**.

(51) The one end **1101** of the flow passage member **110** is fitted into the outlet opening **1212** of the contact member **120** from the other side in the pipe axis direction Ds. The one end **1101** of the flow passage member **110** is disposed at a position recessed from the inlet opening **1211** of the contact member **120** toward the other side in the pipe axis direction Ds. The one end **1101** of the flow passage member **110** does not project from the contact member **120** toward the one side in the pipe axis direction Ds. Thus, the outlet opening **1212** is connected to the flow passage member **110** to provide fluid communication.

(52) The contact member **120** has a contact surface **123** in a region around the inlet opening **1211**. The contact surface **123** extends over the entirety of a region of the contact member **120** excluding a region facing the container **130** at the other side in the pipe axis direction Ds. The contact surface **123** is configured to come into contact with the surface of the target from which the liquid is to be collected. The contact surface **123** projects from the one end **1101** of the flow passage member **110** toward the one side in the pipe axis direction Ds. At least a portion of the contact surface **123** is spherical.

(53) The container **130** receives the liquid discharged from the flow passages **113** at the other end **1102** of the flow passage member **110**. The container **130** includes a peripheral wall **131** and a bottom portion **132** that are integrated together.

(54) The peripheral wall **131** has a tubular shape that extends in the pipe axis direction Ds. In the embodiment of the present disclosure, the peripheral wall **131** has a uniform diameter in the pipe axis direction Ds. The peripheral wall **131** has an inner diameter greater than an outer diameter of the flow passage member **110**. Accordingly, an inner peripheral surface of the peripheral wall **131**

and an outer peripheral surface of the flow passage member **110** are spaced from each other in a radial direction orthogonal to the pipe axis direction **Ds**.

(55) The bottom portion **132** blocks the peripheral wall **131** at the other side in the pipe axis direction **Ds**. The bottom portion **132** is, for example, hemispherical. The bottom portion **132** may also be plate-shaped and extend along a plane substantially orthogonal to the pipe axis direction **Ds**. The container **130** is formed of the peripheral wall **131** and the bottom portion **132** so as to have a tubular shape with a bottom.

(56) The container **130** has an opening **133** at the one side in the pipe axis direction **Ds** (side opposite to the side at which the bottom portion **132** is provided). The contact member **120** is disposed to close the opening **133** of the container **130**.

(57) The bottom portion **132** of the container **130** is spaced from the other end **1102** of the flow passage member **110** in the pipe axis direction **Ds**. Accordingly, a reservoir space **138** that stores the liquid is defined between the bottom portion **132** of the container **130** and the other end **1102** of the flow passage member **110**.

(58) At least a portion or the entirety of the container **130** may be made of a material having transparency characteristics that allow optical observation of the inside of the container **130** from the outside. Examples of the material having transparency characteristics that allow optical observation of the inside of the container **130** from the outside include, but are not limited to, resin materials and glass materials. The material having transparency characteristics that allow optical observation of the inside of the container **130** from the outside may have transparency characteristics such that the amount of the liquid accumulated in the container body is observable visually or with a camera from the outside of the container **130**.

(59) The container **130** has scale marks **139** that indicate an amount of the liquid received in the container **130**.

(60) To collect liquid in the above-described liquid collector **100**, the inlet opening **1211** of the contact member **120** is brought into contact with the surface of the target from which the liquid is to be collected. In some embodiments, the liquid collector **100** collects body fluid (liquid) secreted in a target subject. Examples of a part of the target subject in which the body fluid is secreted include, but are not limited to, an eyeball, regions around the eye at the inner and outer corners of the eye, a part in the oral cavity (for example, a sublingual region), a part in the nasal cavity, and the skin surface. In the embodiment of the present disclosure, the liquid collector **100** collects tear fluid from the outer corner the target subject's eye as the liquid. In this case, as illustrated in FIG. 3, the inlet opening **1211** of the contact member **120** is brought into contact with a surface **1000** of the target subject's face at or around the outer corner of the eye. Accordingly, liquid **999** that is on or near the surface **1000** is introduced through the inlet opening **1211** by capillarity. Thus, the liquid **999** (tear fluid) on or near the surface **1000** is introduced through the inlet opening **1211**.

(61) The liquid **999** introduced through the inlet opening **1211** enters the flow passages **113** at the one end **1101** of the flow passage member **110**, and flows through the flow passages **113** toward the other end **1102**. The liquid **999** is discharged from the flow passages **113** at the other end **1102** of the flow passage member **110**. The liquid **999** discharged from the flow passages **113** is stored in the reservoir space **138** of the container **130**. Thus, the liquid can be efficiently collected.

(62) When the pipe axis direction **Ds** of the liquid collector **100** is inclined with respect to a vertical direction **Dv** upon collection of the liquid **999**, the liquid **999** introduced through the inlet opening **1211** flows downward in the vertical direction **Dv**. The liquid **999** introduced through the inlet opening **1211** tends to flow into a flow passage **113d**, which is one of the flow passages **113** in the flow passage member **110** that is positioned in a lower region in the vertical direction **Dv**. When the liquid **999** flows into the reservoir space **138**, an amount of air corresponding to the volume of the liquid **999** that flows into the reservoir space **138** is pushed out of the reservoir space **138**. The air pushed out of the reservoir space **138** moves toward the inlet opening **1211** through, for example, one or more of the flow passages **113** that are positioned in an upper region in the vertical direction

Dv and through which the liquid 999 does not flow.

(63) FIGS. 4 and 5 illustrate a liquid collector 200 according to an embodiment. As illustrated in FIGS. 4 and 5, the liquid collector 200 includes a flow passage member 210, a contact member 220, and a container 230. The liquid collector 200 collects liquid.

(64) The flow passage member 210 includes an outside tube 211 and an inside member 212. The outside tube 211 has a tubular shape that extends in a pipe axis direction Ds connecting one end 2101 and other end 2102 of the flow passage member 210. In the embodiment of the present disclosure, the outside tube 211 has, for example, a circular shape in cross-section when viewed in the pipe axis direction Ds. The outside tube 211 may also have, for example, an elliptical shape in cross-section when viewed in the pipe axis direction Ds. The outside tube 211 may also have, for example, a polygonal shape, such as a triangular shape, a quadrangular shape, a pentagonal shape, a hexagonal shape, or an octagonal shape in cross-section when viewed in the pipe axis direction Ds.

(65) The inside member 212 is inserted in the outside tube 211. The inside member 212 has a columnar shape that extends in the pipe axis direction Ds. The inside member 212 has a uniform cross-sectional shape in the pipe axis direction Ds. The inside member 212 has a plurality of grooves 2121 that extend in the pipe axis direction Ds. The grooves 2121 are arranged with equal intervals therebetween in a circumferential direction around a central axis of the inside member 212. Each groove 2121 is recessed radially inward from an outer peripheral surface 2122 of the inside member 212. Each groove 2121 extends continuously in the pipe axis direction Ds.

(66) The outer peripheral surface 2122 of the inside member 212 inserted in the outside tube 211 is in contact with, or slightly spaced from, an inner peripheral surface 2111 of the outside tube 211. In the flow passage member 210, flow passages 213 are defined between the inner peripheral surface 2111 of the outside tube 211 and the grooves 2121 in the inside member 212. The flow passages 213 extend in the pipe axis direction Ds to connect the one end 2101 and the other end 2102 of the flow passage member 210.

(67) Thus, the flow passage member 210 includes a plurality of flow passages 213. The flow passages 213 extend in the pipe axis direction Ds connecting the one end 2101 and the other end 2102. The flow passages 213 are substantially parallel to each other.

(68) The contact member 220 is disposed at the one end 2101 of the flow passage member 210. The contact member 220 is configured to come into contact with a target from which the liquid is to be collected. The contact member 220 may include, or may be, an elastic material. When the contact member 220 includes an elastic material, the contact member 220 may be a combination of a portion made of the elastic material and a portion made of another material.

(69) The contact member 220 has a flow-passage-member receiving hole 221. The flow-passage-member receiving hole 221 extends in the pipe axis direction Ds. The flow-passage-member receiving hole 221 has an inlet opening 2211 at an end thereof at one side in the pipe axis direction Ds. The inlet opening 2211 opens at a side opposite to a side at which the flow passage member 210 is disposed in the pipe axis direction Ds. The inlet opening 2211 comes into contact with a surface of the target from which the liquid is to be collected and allows introduction of the liquid that is on or near the surface. The flow-passage-member receiving hole 221 has an outlet opening 2212 at an end thereof at the other side in the pipe axis direction Ds. The outlet opening 2212 opens at the side at which the flow passage member 210 is disposed in the pipe axis direction Ds. The outlet opening 2212 is connected to the inlet opening 2211 to provide fluid communication through the flow-passage-member receiving hole 221.

(70) The one end 2101 of the flow passage member 210 is fitted into the outlet opening 2212 of the contact member 220 from the other side in the pipe axis direction Ds. The one end 2101 of the flow passage member 210 is disposed at a position recessed from the inlet opening 2211 of the contact member 220 toward the other side in the pipe axis direction Ds. The one end 2101 of the flow passage member 210 does not project from the contact member 220 toward the one side in the pipe axis direction Ds. Thus, the outlet opening 2212 is connected to the flow passage member 210 to

provide fluid communication.

(71) The contact member **220** has a contact surface **223** in a region around the inlet opening **2211**. The contact surface **223** is configured to come into contact with the surface of the target from which the liquid is to be collected. The contact surface **223** projects from the one end **2101** of the flow passage member **210** toward the one side in the pipe axis direction **Ds**. At least a portion of the contact surface **223** is spherical. The contact surface **223** is spherical over the entirety of a region of the contact member **220** excluding a region facing the container **230** at the other side in the pipe axis direction **Ds**.

(72) The container **230** receives the liquid discharged from the flow passages **213** at the other end **2102** of the flow passage member **210**. The container **230** includes a peripheral wall **231** and a bottom portion **232** that are integrated together.

(73) The peripheral wall **231** has a tubular shape that extends in the pipe axis direction **Ds**. In the embodiment of the present disclosure, the peripheral wall **231** has a uniform diameter in the pipe axis direction **Ds**. The peripheral wall **231** has an inner diameter substantially equal to an outer diameter of the flow passage member **210**. Accordingly, the other end **2102** of the flow passage member **210** is in contact with an inner peripheral surface **2314** of the peripheral wall **231**. When the other end **2102** of the flow passage member **210** is in contact with the inner peripheral surface **2314** of the peripheral wall **231** as described above, the liquid in the flow passages **213** of the flow passage member **210** easily moves along the inner peripheral surface **2314** of the peripheral wall **231**.

(74) The bottom portion **232** blocks the peripheral wall **231** at the other side in the pipe axis direction **Ds**. The bottom portion **232** is, for example, hemispherical. The bottom portion **232** may also be plate-shaped and extend along a plane substantially orthogonal to the pipe axis direction **Ds**. The container **230** is formed of the peripheral wall **231** and the bottom portion **232** so as to have a tubular shape with a bottom.

(75) The container **230** has an opening **233** at the one side in the pipe axis direction **Ds** (side opposite to the side at which the bottom portion **232** is provided). The contact member **220** is disposed to close the opening **233** of the container **230**.

(76) The bottom portion **232** of the container **230** is spaced from the other end **2102** of the flow passage member **210** in the pipe axis direction **Ds**. Accordingly, a reservoir space **238** that stores the liquid is defined between the bottom portion **232** of the container **230** and the other end **2102** of the flow passage member **210**.

(77) At least a portion or the entirety of the container **230** may be made of a material having transparency characteristics that allow optical observation of the inside of the container **230** from the outside. Examples of the material having transparency characteristics that allow optical observation of the inside of the container **230** from the outside include, but are not limited to, resin materials and glass materials. The material having transparency characteristics that allow optical observation of the inside of the container **230** from the outside may have transparency characteristics such that the amount of the liquid accumulated in the container body is observable visually or with a camera from the outside of the container **230**.

(78) The container **230** has scale marks **239** that indicate an amount of the liquid received in the container **230**.

(79) To collect liquid in the above-described liquid collector **200**, the inlet opening **2211** of the contact member **220** is brought into contact with a surface **1000** of the target from which the liquid is to be collected, as illustrated in FIG. 5. Accordingly, liquid **999** that is on or near the surface **1000** is introduced through the inlet opening **2211** by capillarity. The liquid **999** introduced through the inlet opening **2211** enters the flow passages **213** at the one end **2101** of the flow passage member **210**. The liquid **999** flows toward the other end **2102** through the flow passages **213**. The liquid **999** is discharged from the flow passages **213** at the other end **2102** of the flow passage member **210**. Since the outer peripheral surface **2122** of the flow passage member **210** is in contact

with the inner peripheral surface **2314** of the container **230**, the liquid **999** discharged from the flow passages **213** easily flows along the inner peripheral surface **2314** of the container **230**. The liquid **999** flows downward along the inner peripheral surface **2314**, and is stored in the reservoir space **238** of the container **230**. Thus, the liquid can be efficiently collected.

(80) When the pipe axis direction D_s of the liquid collector **200** is inclined with respect to a vertical direction D_v upon collection of the liquid **999**, the liquid **999** introduced through the inlet opening **2211** flows downward in the vertical direction D_v . The liquid **999** introduced through the inlet opening **2211** tends to flow into a flow passage **213d**, which is one of the flow passages **213** in the flow passage member **210** that is positioned in a lower region in the vertical direction D_v .

(81) When the liquid **999** flows into the reservoir space **238**, an amount of air corresponding to the volume of the liquid **999** that flows into the reservoir space **238** is pushed out of the reservoir space **238**. The air pushed out of the reservoir space **238** flows through one or more of the flow passages **213** that are positioned in an upper region in the vertical direction D_v and through which the liquid **999** does not flow, and is discharged to the outside from the inlet opening **2211**.

(82) FIGS. **6** and **7** illustrate a liquid collector **300** according to an embodiment. As illustrated in FIGS. **6** and **7**, the liquid collector **300** includes a flow passage member **310**, a contact member **320**, a container **330**, and an inner member **340**. The liquid collector **300** collects liquid.

(83) The flow passage member **310** includes an outside tube **311** and an inside member **312**.

(84) The outside tube **311** has a tubular shape that extends in a pipe axis direction D_s connecting one end **3101** and other end **3102** of the flow passage member **310**. In the embodiment of the present disclosure, the outside tube **311** has, for example, a circular shape in cross-section when viewed in the pipe axis direction D_s . The outside tube **311** may also have, for example, an elliptical shape in cross-section when viewed in the pipe axis direction D_s . The outside tube **311** may also have, for example, a polygonal shape, such as a triangular shape, a quadrangular shape, a pentagonal shape, a hexagonal shape, or an octagonal shape in cross-section when viewed in the pipe axis direction D_s .

(85) The inside member **312** is inserted in the outside tube **311**. The inside member **312** has a columnar shape that extends in the pipe axis direction D_s . The inside member **312** has a uniform cross-sectional shape in the pipe axis direction D_s . The inside member **312** has a plurality of grooves **3121** that extend in the pipe axis direction D_s . The grooves **3121** are arranged with equal intervals therebetween in a circumferential direction around a central axis of the inside member **312**. Each groove **3121** is recessed radially inward from an outer peripheral surface **3122** of the inside member **312**. Each groove **3121** extends continuously in the pipe axis direction D_s .

(86) The outer peripheral surface **3122** of the inside member **312** inserted in the outside tube **311** is in contact with, or slightly spaced from, an inner peripheral surface **3111** of the outside tube **311**. In the flow passage member **310**, flow passages **313** are defined between the inner peripheral surface **3111** of the outside tube **311** and the grooves **3121** in the inside member **312**. The flow passages **313** extend in the pipe axis direction D_s to connect the one end **3101** and the other end **3102** of the flow passage member **310**.

(87) Thus, the flow passage member **310** includes a plurality of flow passages **313**. The flow passages **313** extend in the pipe axis direction D_s connecting the one end **3101** and the other end **3102**. The flow passages **313** are substantially parallel to each other.

(88) The contact member **320** is disposed at the one end **3101** of the flow passage member **310**. The contact member **320** is configured to come into contact with a target from which the liquid is to be collected. The contact member **320** may include, or may be, an elastic material. When the contact member **320** includes an elastic material, the contact member **320** may be a combination of a portion made of the elastic material and a portion made of another material.

(89) The contact member **320** has a flow-passage-member receiving hole **321**. The flow-passage-member receiving hole **321** extends in the pipe axis direction D_s . The flow-passage-member receiving hole **321** has an inlet opening **3211** at an end thereof at one side in the pipe axis direction

Ds. The inlet opening **3211** opens at a side opposite to a side at which the flow passage member **310** is disposed in the pipe axis direction Ds. The inlet opening **3211** comes into contact with a surface of the target from which the liquid is to be collected and allows introduction of the liquid that is on or near the surface.

(90) The flow-passage-member receiving hole **321** has an outlet opening **3212** at an end thereof at the other side in the pipe axis direction Ds. The outlet opening **3212** opens at the side at which the flow passage member **310** is disposed in the pipe axis direction Ds. The outlet opening **3212** is connected to the inlet opening **3211** to provide fluid communication through the flow-passage-member receiving hole **321**.

(91) The one end **3101** of the flow passage member **310** is fitted into the outlet opening **3212** of the contact member **320** from the other side in the pipe axis direction Ds. The one end **3101** of the flow passage member **310** is disposed at a position recessed from the inlet opening **3211** of the contact member **320** toward the other side in the pipe axis direction Ds. The one end **3101** of the flow passage member **310** does not project from the contact member **320** toward the one side in the pipe axis direction Ds. Thus, the outlet opening **3212** is connected to the flow passage member **310** to provide fluid communication.

(92) The contact member **320** has a contact surface **323** in a region around the inlet opening **3211**. The contact surface **323** is configured to come into contact with the surface of the target from which the liquid is to be collected. The contact surface **323** projects from the one end **3101** of the flow passage member **310** toward the one side in the pipe axis direction Ds. At least a portion of the contact surface **323** is spherical. The contact surface **323** is spherical over the entirety of a region of the contact member **320** excluding a region facing the container **330** at the other side in the pipe axis direction Ds.

(93) The container **330** receives the liquid discharged from the flow passages **313** at the other end **3102** of the flow passage member **310**. The container **330** includes a peripheral wall **331** and a bottom portion **332** that are integrated together.

(94) The peripheral wall **331** has a tubular shape that extends in the pipe axis direction Ds. In the embodiment of the present disclosure, the peripheral wall **331** has an inner diameter that gradually decreases toward the bottom portion **332** at the other side from the one side in the pipe axis direction Ds.

(95) The bottom portion **332** blocks the peripheral wall **331** at the other side in the pipe axis direction Ds. The bottom portion **332** is, for example, hemispherical. The bottom portion **332** may also be plate-shaped and extend along a plane substantially orthogonal to the pipe axis direction Ds. The container **330** is formed of the peripheral wall **331** and the bottom portion **332** so as to have a tubular shape with a bottom.

(96) The container **330** has an opening **333** at the one side in the pipe axis direction Ds (side opposite to the side at which the bottom portion **332** is provided). The contact member **320** is disposed to close the opening **333** of the container **330**.

(97) The bottom portion **332** of the container **330** is spaced from the other end **3102** of the flow passage member **310** in the pipe axis direction Ds. Accordingly, a reservoir space **338** that stores the liquid is defined between the bottom portion **332** of the container **330** and the other end **3102** of the flow passage member **310**.

(98) At least a portion or the entirety of the container **330** may be made of a material having transparency characteristics that allow optical observation of the inside of the container **330** from the outside. Examples of the material having transparency characteristics that allow optical observation of the inside of the container **330** from the outside include, but are not limited to, resin materials and glass materials. The material having transparency characteristics that allow optical observation of the inside of the container **330** from the outside may have transparency characteristics such that the amount of the liquid accumulated in the container body is observable visually or with a camera from the outside of the container **330**.

(99) The container **330** has scale marks **339** that indicate an amount of the liquid received in the container **330**.

(100) The inner member **340** is disposed in the container **330**. The inner member **340** includes a cone-shaped portion **341** having an inner diameter that gradually decreases toward the bottom portion **332** in the pipe axis direction **Ds**. In the embodiment of the present disclosure, the cone-shaped portion **341** is formed over the entire region of the inner member **340** in the pipe axis direction **Ds**. In some embodiments, the cone-shaped portion **341** may be formed over only a partial region of the inner member **340** in the pipe axis direction **Ds**. The cone-shaped portion **341** has a first opening **3411** at an end adjacent to the opening **333**, and an inner diameter of the first opening **3411** is greater than an outer diameter of the flow passage member **310** at the other end **3102**. The cone-shaped portion **341** has a second opening **3412** at an end adjacent to the bottom portion **332**, and an inner diameter of the second opening **3412** is less than the outer diameter of the flow passage member **310** at the other end **3102**. Accordingly, the other end **3102** of the flow passage member **310** is in contact with an inner peripheral surface of the cone-shaped portion **341** at a position between the first opening **3411** and the second opening **3412** of the cone-shaped portion **341**. When the other end **3102** of the flow passage member **310** is in contact with the cone-shaped portion **341** as described above, the liquid in the flow passages **313** of the flow passage member **310** easily flows along the inner peripheral surface of the cone-shaped portion **341**.

(101) To collect liquid in the above-described liquid collector **300**, the inlet opening **3211** of the contact member **320** is brought into contact with a surface **1000** of the target from which the liquid is to be collected, as illustrated in FIG. 7. Accordingly, liquid **999** that is on or near the surface **1000** is introduced through the inlet opening **3211** by capillarity. The liquid **999** introduced through the inlet opening **3211** enters the flow passages **313** at the one end **3101** of the flow passage member **310**. The liquid **999** flows toward the other end **3102** through the flow passages **313**. The liquid **999** is discharged from the flow passages **313** at the other end **3102** of the flow passage member **310**. The liquid **999** discharged from the flow passages **313** flows downward along the inner peripheral surface of the cone-shaped portion **341**. The liquid **999** falls from the second opening **3412** of the cone-shaped portion **341**, and is stored in the reservoir space **338** of the container **330**. Thus, the liquid can be efficiently collected.

(102) FIGS. 8 and 9 illustrate a liquid collector **400** according to an embodiment. As illustrated in FIGS. 8 and 9, the liquid collector **400** includes a flow passage member **410**, a contact member **420**, a container **430**, and an inner member **440**. The liquid collector **400** collects liquid.

(103) The flow passage member **410** includes an outside tube **411** and an inside member **412**.

(104) The outside tube **411** has a tubular shape that extends in a pipe axis direction **Ds** connecting one end **4101** and other end **4102** of the flow passage member **410**. In the embodiment of the present disclosure, the outside tube **411** has, for example, a circular shape in cross-section when viewed in the pipe axis direction **Ds**. The outside tube **411** may also have, for example, an elliptical shape in cross-section when viewed in the pipe axis direction **Ds**. The outside tube **411** may also have, for example, a polygonal shape, such as a triangular shape, a quadrangular shape, a pentagonal shape, a hexagonal shape, or an octagonal shape in cross-section when viewed in the pipe axis direction **Ds**.

(105) The inside member **412** is inserted in the outside tube **411**. The inside member **412** has a columnar shape that extends in the pipe axis direction **Ds**. The inside member **412** has a uniform cross-sectional shape in the pipe axis direction **Ds**. The inside member **412** has a plurality of grooves **4121** that extend in the pipe axis direction **Ds**. The grooves **4121** are arranged with equal intervals therebetween in a circumferential direction around a central axis of the inside member **412**. Each groove **4121** is recessed radially inward from an outer peripheral surface **4122** of the inside member **412**. Each groove **4121** extends continuously in the pipe axis direction **Ds**.

(106) The outer peripheral surface **4122** of the inside member **412** inserted in the outside tube **411** is in contact with, or slightly spaced from, an inner peripheral surface **4111** of the outside tube **411**.

In the flow passage member **410**, flow passages **413** are defined between the inner peripheral surface **4111** of the outside tube **411** and the grooves **4121** in the inside member **412**. The flow passages **413** extend in the pipe axis direction **Ds** to connect the one end **4101** and the other end **4102** of the flow passage member **410**.

(107) Thus, the flow passage member **410** includes a plurality of flow passages **413**. The flow passages **413** extend in the pipe axis direction **Ds** connecting the one end **4101** and the other end **4102**. The flow passages **413** are substantially parallel to each other.

(108) The contact member **420** is disposed at the one end **4101** of the flow passage member **410**. The contact member **420** is configured to come into contact with a target from which the liquid is to be collected. The contact member **420** may include, or may be, an elastic material. When the contact member **420** includes an elastic material, the contact member **420** may be a combination of a portion made of the elastic material and a portion made of another material.

(109) The contact member **420** has a flow-passage-member receiving hole **421**. The flow-passage-member receiving hole **421** extends in the pipe axis direction **Ds**. The flow-passage-member receiving hole **421** has an inlet opening **4211** at an end thereof at one side in the pipe axis direction **Ds**. The inlet opening **4211** opens at a side opposite to a side at which the flow passage member **410** is disposed in the pipe axis direction **Ds**. The inlet opening **4211** comes into contact with a surface of the target from which the liquid is to be collected and allows introduction of the liquid that is on or near the surface.

(110) The contact member **420** includes a sleeve **425** that is integral therewith. The sleeve **425** projects from the contact member **420** toward the other end **4102** of the flow passage member **410**. The sleeve **425** extends in the pipe axis direction **Ds**. The sleeve **425** is inserted in the container **430**.

(111) The flow-passage-member receiving hole **421** has an outlet opening **4212** at an end thereof at the other side in the pipe axis direction **Ds**. In the embodiment of the present disclosure, the outlet opening **4212** is formed in the sleeve **425**. The outlet opening **4212** opens at the side at which the flow passage member **410** is disposed in the pipe axis direction **Ds**. The outlet opening **4212** is connected to the inlet opening **4211** to provide fluid communication through the flow-passage-member receiving hole **421**.

(112) The one end **4101** of the flow passage member **410** is fitted into the outlet opening **4212** of the contact member **420** from the other side in the pipe axis direction **Ds**. Since the contact member **420** includes the sleeve **425**, the one end **4101** of the flow passage member **410** is retained by the contact member **420** (including the sleeve **425**) over a longer length in the pipe axis direction **Ds**. Accordingly, the flow passage member **410** is more strongly supported. Thus, the outlet opening **4212** is connected to the flow passage member **410** to provide fluid communication.

(113) The one end **4101** of the flow passage member **410** is disposed at a position recessed from the inlet opening **4211** of the contact member **420** toward the other side in the pipe axis direction **Ds**. The one end **4101** of the flow passage member **410** does not project from the contact member **420** toward the one side in the pipe axis direction **Ds**.

(114) The contact member **420** has a contact surface **423** in a region around the inlet opening **4211**. The contact surface **423** is configured to come into contact with the surface of the target from which the liquid is to be collected. The contact surface **423** projects from the one end **4101** of the flow passage member **410** toward the one side in the pipe axis direction **Ds**. At least a portion of the contact surface **423** is spherical. The contact surface **423** is spherical over the entirety of a region of the contact member **420** excluding a region facing the container **430** at the other side in the pipe axis direction **Ds**.

(115) The container **430** receives the liquid discharged from the flow passages **413** at the other end **4102** of the flow passage member **410**. The container **430** includes a peripheral wall **431** and a bottom portion **432** that are integrated together.

(116) The peripheral wall **431** has a tubular shape that extends in the pipe axis direction **Ds**. In the

embodiment of the present disclosure, the peripheral wall **431** has an inner diameter that gradually decreases toward the bottom portion **432** at the other side from the one side in the pipe axis direction **Ds**.

(117) The bottom portion **432** blocks the peripheral wall **431** at the other side in the pipe axis direction **Ds**. The bottom portion **432** is, for example, hemispherical. The bottom portion **432** may also be plate-shaped and extend along a plane substantially orthogonal to the pipe axis direction **Ds**. The container **430** is formed of the peripheral wall **431** and the bottom portion **432** so as to have a tubular shape with a bottom.

(118) The container **430** has an opening **433** at the one side in the pipe axis direction **Ds** (side opposite to the side at which the bottom portion **432** is provided). The contact member **420** is disposed to close the opening **433** of the container **430**.

(119) The bottom portion **432** of the container **430** is spaced from the other end **4102** of the flow passage member **410** in the pipe axis direction **Ds**. Accordingly, a reservoir space **438** that stores the liquid is defined between the bottom portion **432** of the container **430** and the other end **4102** of the flow passage member **410**.

(120) At least a portion or the entirety of the container **430** may be made of a material having transparency characteristics that allow optical observation of the inside of the container **430** from the outside. Examples of the material having transparency characteristics that allow optical observation of the inside of the container **430** from the outside include, but are not limited to, resin materials and glass materials. The material having transparency characteristics that allow optical observation of the inside of the container **430** from the outside may have transparency characteristics such that the amount of the liquid accumulated in the container body is observable visually or with a camera from the outside of the container **430**.

(121) The container **430** has scale marks **439** that indicate an amount of the liquid received in the container **430**.

(122) The inner member **440** is disposed in the container **430**. The inner member **440** includes a cone-shaped portion **441** having an inner diameter that gradually decreases toward the bottom portion **432** in the pipe axis direction **Ds**. In the embodiment of the present disclosure, the cone-shaped portion **441** is formed over the entire region of the inner member **440** in the pipe axis direction **Ds**. In some embodiments, the cone-shaped portion **441** may be formed over only a partial region of the inner member **440** in the pipe axis direction **Ds**. The cone-shaped portion **441** has a first opening **4411** at an end adjacent to the opening **433**, and an inner diameter of the first opening **4411** is greater than an outer diameter of the flow passage member **410** at the other end **4102**. The cone-shaped portion **441** has a second opening **4412** at an end adjacent to the bottom portion **432**, and an inner diameter of the second opening **4412** is less than the outer diameter of the flow passage member **410** at the other end **4102**. Accordingly, the other end **4102** of the flow passage member **410** is in contact with an inner peripheral surface of the cone-shaped portion **441** at a position between the first opening **4411** and the second opening **4412** of the cone-shaped portion **441**. When the other end **4102** of the flow passage member **410** is in contact with the cone-shaped portion **441** as described above, the liquid in the flow passages **413** of the flow passage member **410** easily flows along the cone-shaped portion **441**.

(123) To collect liquid in the above-described liquid collector **400**, the inlet opening **4211** of the contact member **420** is brought into contact with a surface **1000** of the target from which the liquid is to be collected, as illustrated in FIG. 9. Accordingly, liquid **999** that is on or near the surface **1000** is introduced through the inlet opening **4211** by capillarity. The liquid **999** introduced through the inlet opening **4211** enters the flow passages **413** at the one end **4101** of the flow passage member **410**. The liquid **999** flows toward the other end **4102** through the flow passages **413**. The liquid **999** is discharged from the flow passages **413** at the other end **4102** of the flow passage member **410**. The liquid **999** discharged from the flow passages **413** flows downward along the inner peripheral surface of the cone-shaped portion **441**. The liquid **999** falls from the second

opening **441** of the cone-shaped portion **441**, and is stored in the reservoir space **438** of the container **430**. Thus, the liquid can be efficiently collected.

(124) FIGS. **10** and **11** illustrate a liquid collector **500** according to an embodiment. As illustrated in FIGS. **10** and **11**, the liquid collector **500** includes a flow passage member **510**, a container **530**, and an inner member **540**. The liquid collector **500** collects liquid.

(125) The flow passage member **510** includes an outside tube **511** and an inside member **512**.

(126) The outside tube **511** has a tubular shape that extends in a pipe axis direction **Ds** connecting one end **5101** and other end **5102** of the flow passage member **510**. In the embodiment of the present disclosure, the outside tube **511** has, for example, a circular shape in cross-section when viewed in the pipe axis direction **Ds**. The outside tube **511** may also have, for example, an elliptical shape in cross-section when viewed in the pipe axis direction **Ds**. The outside tube **511** may also have, for example, a polygonal shape, such as a triangular shape, a quadrangular shape, a pentagonal shape, a hexagonal shape, or an octagonal shape in cross-section when viewed in the pipe axis direction **Ds**.

(127) The inside member **512** is inserted in the outside tube **511**. The inside member **512** has a columnar shape that extends in the pipe axis direction **Ds**. The inside member **512** has a uniform cross-sectional shape in the pipe axis direction **Ds**. The inside member **512** has a plurality of grooves **5121** that extend in the pipe axis direction **Ds**. The grooves **5121** are arranged with equal intervals therebetween in a circumferential direction around a central axis of the inside member **512**. Each groove **5121** is recessed radially inward from an outer peripheral surface **5122** of the inside member **512**. Each groove **5121** extends continuously in the pipe axis direction **Ds**.

(128) The outer peripheral surface **5122** of the inside member **512** inserted in the outside tube **511** is in contact with, or slightly spaced from, an inner peripheral surface **5111** of the outside tube **511**. In the flow passage member **510**, flow passages **513** are defined between the inner peripheral surface **5111** of the outside tube **511** and the grooves **5121** in the inside member **512**. The flow passages **513** extend in the pipe axis direction **Ds** to connect the one end **5101** and the other end **5102** of the flow passage member **510**.

(129) Thus, the flow passage member **510** includes a plurality of flow passages **513**. The flow passages **513** extend in the pipe axis direction **Ds** connecting the one end **5101** and the other end **5102**. The flow passages **513** are substantially parallel to each other.

(130) The container **530** receives the liquid discharged from the flow passages **513** at the other end **5102** of the flow passage member **510**. The container **530** includes a peripheral wall **531** and a bottom portion **532** that are integrated together.

(131) The peripheral wall **531** has a tubular shape that extends in the pipe axis direction **Ds**. In the embodiment of the present disclosure, the peripheral wall **531** has an inner diameter that gradually decreases toward the bottom portion **532** at the other side from the one side in the pipe axis direction **Ds**.

(132) The bottom portion **532** blocks the peripheral wall **531** at the other side in the pipe axis direction **Ds**. The bottom portion **532** is, for example, hemispherical. The bottom portion **532** may also be plate-shaped and extend along a plane substantially orthogonal to the pipe axis direction **Ds**. The container **530** is formed of the peripheral wall **531** and the bottom portion **532** so as to have a tubular shape with a bottom.

(133) The container **530** has an opening **533** at one side in the pipe axis direction **Ds** (side opposite to the side at which the bottom portion **532** is provided). The flow passage member **510** is inserted into the container **530** through the opening **533**. The bottom portion **532** of the container **530** is spaced from the other end **5102** of the flow passage member **510** in the pipe axis direction **Ds**. Accordingly, a reservoir space **538** that stores the liquid is defined between the bottom portion **532** of the container **530** and the other end **5102** of the flow passage member **510**.

(134) At least a portion or the entirety of the container **530** may be made of a material having transparency characteristics that allow optical observation of the inside of the container **530** from

the outside. Examples of the material having transparency characteristics that allow optical observation of the inside of the container **530** from the outside include, but are not limited to, resin materials and glass materials. The material having transparency characteristics that allow optical observation of the inside of the container **530** from the outside may have transparency characteristics such that the amount of the liquid accumulated in the container body is observable visually or with a camera from the outside of the container **530**.

(135) The container **530** has scale marks **539** that indicate an amount of the liquid received in the container **530**.

(136) The inner member **540** is disposed in the container **530**. The inner member **540** includes a cone-shaped portion **541** having an inner diameter that gradually decreases toward the bottom portion **532** in the pipe axis direction **Ds**. In the embodiment of the present disclosure, the cone-shaped portion **541** is formed over the entire region of the inner member **540** in the pipe axis direction **Ds**. In some embodiments, the cone-shaped portion **541** may be formed over only a partial region of the inner member **540** in the pipe axis direction **Ds**. The cone-shaped portion **541** has a first opening **5411** at an end adjacent to the opening **533**, and an inner diameter of the first opening **5411** is greater than an outer diameter of the flow passage member **510** at the other end **5102**. The cone-shaped portion **541** has a second opening **5412** at an end adjacent to the bottom portion **532**, and an inner diameter of the second opening **5412** is less than the outer diameter of the flow passage member **510** at the other end **5102**. Accordingly, the other end **5102** of the flow passage member **510** is in contact with an inner peripheral surface of the cone-shaped portion **541** at a position between the first opening **5411** and the second opening **5412** of the cone-shaped portion **541**. When the other end **5102** of the flow passage member **510** is in contact with the cone-shaped portion **541** as described above, the liquid in the flow passages **513** of the flow passage member **510** easily flows along the cone-shaped portion **541**.

(137) To collect liquid in the above-described liquid collector **500**, the one end **5101** of the flow passage member **510** is brought into contact with a surface **1000** of the target from which the liquid is to be collected, as illustrated in FIG. **11**. The one end **5101** may have a hydrophilic or embossed surface. Accordingly, liquid **999** that is on or near the surface **1000** is introduced from the one end **5101** of the flow passage member **510** by capillarity. The liquid **999** introduced into the flow passage member **510** flows toward the other end **5102** through the flow passages **513**. The liquid **999** is discharged from the flow passages **513** at the other end **5102** of the flow passage member **510**. The liquid **999** discharged from the flow passages **513** flows downward along the inner peripheral surface of the cone-shaped portion **541**. The liquid **999** falls from the second opening **5412** of the cone-shaped portion **541**, and is stored in the reservoir space **538** of the container **530**. Thus, the liquid can be efficiently collected.

(138) FIG. **12** illustrates a flow passage member **710** of a liquid collector according to an embodiment. Referring to FIG. **12**, the flow passage member **710** has a columnar shape that extends in a pipe axis direction **Ds** connecting one end and other end of the flow passage member **710** (direction orthogonal to the plane of FIG. **12**). In the embodiment of the present disclosure, the flow passage member **710** has, for example, a circular shape in cross-section when viewed in the pipe axis direction **Ds**. The flow passage member **710** may also have, for example, an elliptical shape in cross-section when viewed in the pipe axis direction **Ds**. The flow passage member **710** may also have, for example, a polygonal shape, such as a triangular shape, a quadrangular shape, a pentagonal shape, a hexagonal shape, or an octagonal shape in cross-section when viewed in the pipe axis direction **Ds**.

(139) The flow passage member **710** has a plurality of through holes **715** that extend therethrough in the pipe axis direction **Ds** to connect one end and the other end of the flow passage member **710**. In the embodiment of the present disclosure, a through hole **715A**, which is one of the through holes **715**, is disposed at the center of the flow passage member **710**. Through holes **715B**, which are the remaining ones of the through holes **715**, are arranged in an outer peripheral region of the

flow passage member **710** with equal intervals therebetween in the circumferential direction. The through holes **715** are substantially parallel to the pipe axis direction **Ds**.

(140) Each through hole **715** serves as a flow passage **713** for the liquid in the flow passage member **710**. The flow passage member **710** includes a plurality of flow passages **713** (through holes **715**). The flow passages **713** extend in the pipe axis direction **Ds** connecting the one end and the other end. The flow passages **713** are substantially parallel to each other.

(141) In some embodiments, the above-described flow passage member **710** may be used instead of the flow passage members **110**, **210**, **310**, **410**, and **510**.

(142) FIG. **13** illustrates a flow passage member **810** of a liquid collector according to an embodiment. As illustrated in FIG. **13**, the flow passage member **810** includes a plurality of single pipes **812**.

(143) Each single pipe **812** has a tubular shape that extends in a pipe axis direction **Ds** (direction orthogonal to the plane of FIG. **13**). Each single pipe **812** has a uniform cross-sectional shape in the pipe axis direction **Ds**. In the embodiment of the present disclosure, each single pipe **812** has, for example, a circular shape in cross-section when viewed in the pipe axis direction **Ds**. The flow passage member **810** is formed by bundling the single pipes **812**. The single pipes **812** may be bundled and joined together by, for example, without limitation, adhesion, fusion bonding, or welding. The single pipes **812** may be bundled together by using, for example, without limitation, a belt-shaped band or a ring. In some embodiments, each single pipe may have a cross-sectional shape that varies in the pipe axis direction. For example, each single pipe may be a combination of two single pipes having different sizes (for example, hole diameters).

(144) A single pipe **812A**, which is one of the single pipes **812**, is disposed at the center of the flow passage member **810**. Single pipes **812B**, which are the remaining ones of the single pipes **812**, are arranged in contact with each other in the circumferential direction in a region radially outside the single pipe **812A**. Each single pipe **812** extends continuously in the pipe axis direction **Ds**. Each single pipe **812** has a hole **815** that extends therethrough in the pipe axis direction **Ds**.

(145) A plurality of gaps **816** are formed between the single pipe **812A** and the single pipes **812B** arranged radially outside the single pipe **812A**. Each gap **816** extends continuously in the pipe axis direction **Ds** to connect one end and the other end of the flow passage member **810**.

(146) Thus, the holes **815** and the gaps **816** are defined in the flow passage member **810**. The holes **815** and the gaps **816** each serve as a flow passage **813** for the liquid in the flow passage member **810**. The flow passage member **810** has a plurality of flow passages **813** (holes **815** and gaps **816**). The flow passages **813** extend in the pipe axis direction **Ds** connecting the one end and the other end of the flow passage member **810**. The flow passages **813** are substantially parallel to each other.

(147) In some embodiments, the above-described flow passage member **810** may be used instead of the flow passage members **110**, **210**, **310**, **410**, and **510**.

(148) FIG. **14** illustrates a flow passage member **910** of a liquid collector according to an embodiment. Referring to FIG. **14**, the flow passage member **910** includes an outer tube **911** and inner pipes **912**.

(149) The outer tube **911** has a tubular shape that extends in a pipe axis direction **Ds** connecting one end and other end of the flow passage member **910** (direction orthogonal to the plane of FIG. **14**). In the embodiment of the present disclosure, the outer tube **911** has, for example, a circular shape in cross-section when viewed in the pipe axis direction **Ds**.

(150) The inner pipes **912** are inserted in the outer tube **911**. Each inner pipe **912** has a tubular shape that extends in the pipe axis direction **Ds**. Each inner pipe **912** has a uniform cross-sectional shape in the pipe axis direction **Ds**. In the embodiment of the present disclosure, each inner pipe **912** has, for example, a circular shape in cross-section when viewed in the pipe axis direction **Ds**. The inner pipes **912** have an outer diameter less than an inner diameter of the outer tube **911**. The inner pipes **912** are inserted in the outer tube **911**. An inner pipe **912A**, which is one of the inner

pipes **912**, is disposed at the center of the outer tube **911**. Inner pipes **912B**, which are the remaining ones of the inner pipes **912**, are arranged in contact with each other in the circumferential direction in a region radially outside the inner pipe **912A**. Each inner pipe **912** extends continuously in the pipe axis direction **Ds**. Each inner pipe **912** has a hole **915** that extends therethrough in the pipe axis direction **Ds**. In some embodiments, each single pipe may have a cross-sectional shape that varies in the pipe axis direction. For example, each single pipe may be a combination of two single pipes having different sizes (for example, hole diameters).

(151) A plurality of gaps **916** are formed between the inner pipe **912A** and the inner pipes **912B** arranged radially outside the inner pipe **912A**. Each gap **916** extends continuously in the pipe axis direction **Ds** to connect the one end and the other end of the flow passage member **910**.

(152) The inner pipes **912B** inserted in the outer tube **911** have outer peripheral surfaces **9122** that are in contact with, or slightly spaced from, an inner peripheral surface **9111** of the outer tube **911**. In the flow passage member **910**, the inner peripheral surface **9111** of the outer tube **911** and the outer peripheral surfaces **9122** of two of the inner pipes **912B** that are adjacent to each other in the circumferential direction define an outer peripheral gap **917** therebetween. The outer peripheral gap **917** continuously extends in the pipe axis direction **Ds** to connect one end and the other end of the flow passage member **910**.

(153) Thus, the holes **915**, the gaps **916**, and the outer peripheral gaps **917** are defined in the flow passage member **910**. The holes **915**, the gaps **916**, and the outer peripheral gaps **917** each serve as a flow passage **913** for the liquid in the flow passage member **910**. The flow passage member **910** has a plurality of flow passages **913** (holes **915**, gaps **916**, and outer peripheral gaps **917**). The flow passages **913** extend in the pipe axis direction **Ds** connecting the one end and the other end of the flow passage member **910**. The flow passages **913** are substantially parallel to each other.

(154) In some embodiments, the above-described flow passage member **910** may be used instead of the flow passage members **110**, **210**, **310**, **410**, and **510**.

(155) FIG. **15** illustrates a flow passage member **610** of a liquid collector according to an embodiment. Referring to FIG. **15**, the flow passage member **610** has a columnar shape that extends in a pipe axis direction **Ds** connecting one end and other end of the flow passage member **610** (direction orthogonal to the plane of FIG. **15**). The flow passage member **610** has a uniform cross-sectional shape in the pipe axis direction **Ds**. The flow passage member **610** has a plurality of grooves **615**. The grooves **615** are arranged with equal intervals therebetween in a circumferential direction around a central axis of the flow passage member **610**. Each groove **615** is recessed radially inward from an outer peripheral surface **6122** of the flow passage member **610**. Each groove **615** extends continuously in the pipe axis direction **Ds**.

(156) Each groove **615** serves as a flow passage **613** for the liquid in the flow passage member **610**. The flow passage member **610** includes a plurality of flow passages **613** (grooves **615**). The flow passages **613** extend in the pipe axis direction **Ds** connecting the one end and the other end of the flow passage member **610**. The flow passages **613** are substantially parallel to each other.

(157) In some embodiments, the above-described flow passage member **610** may be used instead of the flow passage members **110**, **210**, **310**, **410**, and **510**.

(158) FIG. **16** illustrates a flow passage member **880** of a liquid collector according to an embodiment. As illustrated in FIG. **16**, the flow passage member **880** includes one pipe **882**.

(159) The pipe **882** has a tubular shape that extends in a pipe axis direction **Ds** (direction orthogonal to the plane of FIG. **16**). The pipe **882** has a uniform cross-sectional shape in the pipe axis direction **Ds**. In the embodiment of the present disclosure, the pipe **882** has, for example, a circular shape in cross-section when viewed in the pipe axis direction **Ds**. The pipe **882** may also have, for example, an elliptical shape in cross-section when viewed in the pipe axis direction **Ds**. The pipe **882** may also have, for example, a polygonal shape, such as a triangular shape, a quadrangular shape, a pentagonal shape, a hexagonal shape, or an octagonal shape in cross-section when viewed in the pipe axis direction **Ds**. The pipe **882** has a hole **885** that extends therethrough

in the pipe axis direction Ds.

(160) The hole **885** serves as a flow passage **883** for the liquid in the flow passage member **880**. The flow passage member **880** includes the flow passage **883** (hole **885**). The flow passage **883** extends in the pipe axis direction Ds connecting one end and other end of the flow passage member **880**.

(161) In some embodiments, the above-described flow passage member **880** may be used instead of the flow passage members **110**, **210**, **310**, **410**, and **510**.

(162) The present disclosure also provides the following embodiments:

(163) A001

(164) A liquid collector that collects liquid, the liquid collector including: a flow passage member including a flow passage that connects one end and other end of the flow passage member; a contact member that is disposed at the one end of the flow passage member, that includes an elastic material, and that is configured to come into contact with a target from which liquid is to be collected; and a container that receives the liquid discharged from the flow passage at the other end of the flow passage member.

A101

(165) The liquid collector according to Embodiment A001, wherein the flow passage member includes a plurality of the flow passages that extend in a pipe axis direction connecting the one end and the other end of the flow passage member.

A102

(166) The liquid collector according to Embodiment A101, wherein the plurality of the flow passages are substantially parallel to each other.

A103

(167) The liquid collector according to Embodiment A101 or A102, wherein the flow passage member has a columnar shape that extends in the pipe axis direction connecting the one end and the other end of the flow passage member, and has a plurality of grooves that extend in the pipe axis direction along an outer peripheral surface of the flow passage member.

A104

(168) The liquid collector according to Embodiment A101 or A102, wherein the flow passage member includes an outside tube having a tubular shape that extends in the pipe axis direction, and an inside member inserted in the outside tube and having a plurality of grooves that extend in the pipe axis direction along an outer peripheral surface of the flow passage member, and wherein an inner peripheral surface of the outside tube and the plurality of grooves in the inside member define the plurality of the flow passages.

A105

(169) The liquid collector according to Embodiment A101 or A102, wherein the flow passage member has a columnar shape that extends in the pipe axis direction, and the plurality of the flow passages extend through the flow passage member in the pipe axis direction to provide communication between the one end and the other end of the flow passage member.

A106

(170) The liquid collector according to Embodiment A101 or A102, wherein the flow passage member includes a plurality of single pipes that are bundled together and that extend in the pipe axis direction.

A107

(171) The liquid collector according to Embodiment A101 or A102, wherein the flow passage member includes an outer tube having a tubular shape that extends in the pipe axis direction, and a plurality of inner pipes that are inserted in the outer tube and that extend in the pipe axis direction.

A111

(172) The liquid collector according to any one of Embodiments A001 to A107, wherein the container includes a peripheral wall having a tubular shape that extends in the pipe axis direction,

and a bottom portion that blocks the peripheral wall at one side in the pipe axis direction.

A112

(173) The liquid collector according to Embodiment A111, wherein the bottom portion of the container is spaced from the other end of the flow passage member in the pipe axis direction, and wherein a reservoir space that stores the liquid is defined between the bottom portion of the container and the other end of the flow passage member.

A113

(174) The liquid collector according to Embodiment A111 or A112, wherein the peripheral wall of the container has an inner diameter that gradually decreases toward the bottom portion in the pipe axis direction.

A114

(175) The liquid collector according to any one of Embodiments A111 to A113, wherein the other end of the flow passage member is in contact with an inner peripheral surface of the peripheral wall of the container.

A115

(176) The liquid collector according to any one of Embodiments A111 to A114, wherein the container has an air hole that provides communication with an outside of the container.

A116

(177) The liquid collector according to any one of Embodiments A111 to A115, wherein the container has a scale mark that indicates an amount of the liquid received in the

A121

(178) The liquid collector according to any one of Embodiments A101 to A115, further including: a cone-shaped portion disposed in the container, wherein at least a portion of the cone-shaped portion in the pipe axis direction has an inner diameter that gradually decreases toward the bottom portion in the pipe axis direction.

A122

(179) The liquid collector according to Embodiment A121, wherein the other end of the flow passage member is in contact with an inner peripheral surface of the cone-shaped portion.

A131

(180) The liquid collector according to any one of Embodiments A001 to A122, wherein the contact member includes an inlet opening that comes into contact with a surface of the target from which the liquid is to be collected and allows introduction of the liquid that is on or near the surface, and an outlet opening that is connected to the inlet opening to provide fluid communication and also connected to the flow passage member to provide fluid communication.

A132

(181) The liquid collector according to Embodiment A131, wherein the one end of the flow passage member is fitted to the outlet opening of the contact member.

A133

(182) The liquid collector according to Embodiment A132, wherein the contact member includes a sleeve that has a tubular shape and extends toward the other end of the flow passage member.

A134

(183) The liquid collector according to any one of Embodiments A131 to A133, wherein the contact member has a contact surface in a region around the inlet opening, the contact surface coming into contact with the surface of the target from which the liquid is to be collected.

A135

(184) The liquid collector according to Embodiment A134, wherein at least a portion of the contact surface is spherical.

A136

(185) The liquid collector according to any one of Embodiments A101 to A135, wherein the container has an opening at a side opposite to a side at which the bottom portion is provided in the

pipe axis direction, and wherein the contact member is disposed to close the opening.

A002

(186) A liquid collector that collects liquid, the liquid collector including: a flow passage member including a plurality of flow passages that connect one end and other end of the flow passage member; and a container that receives the liquid discharged from the plurality of flow passages at the other end of the flow passage member.

A201

(187) The liquid collector according to Embodiment A002, further including: a cone-shaped portion disposed in the container, wherein at least a portion of the cone-shaped portion in the pipe axis direction has an inner diameter that gradually decreases toward the bottom portion in the pipe axis direction.

(188) While preferred embodiments of the present invention have been shown and described herein, it will be obvious to those skilled in the art that such embodiments are provided by way of example only. It is not intended that the invention be limited by the specific examples provided within the specification. While the invention has been described with reference to the aforementioned specification, the descriptions and illustrations of the embodiments herein are not meant to be construed in a limiting sense. Numerous variations, changes, and substitutions will now occur to those skilled in the art without departing from the invention. Furthermore, it shall be understood that all aspects of the invention are not limited to the specific depictions, configurations or relative proportions set forth herein which depend upon a variety of conditions and variables. It should be understood that various alternatives to the embodiments of the invention described herein may be employed in practicing the invention. It is therefore contemplated that the invention shall also cover any such alternatives, modifications, variations or equivalents. It is intended that the following claims define the scope of the invention and that methods and structures within the scope of these claims and their equivalents be covered thereby.

REFERENCE SIGNS LIST

(189) **100** liquid collector **110** flow passage member **111** outside tube **112** inside member **113** flow passage **113d** flow passage **120** contact member **121** flow-passage-member receiving hole **123** contact surface **130** container **131** peripheral wall **132** bottom portion **133** opening **138** reservoir space **139** scale mark **200** liquid collector **210** flow passage member **211** outside tube **212** inside member **213** flow passage **213d** flow passage **220** contact member **221** flow-passage-member receiving hole **223** contact surface **230** container **231** peripheral wall **232** bottom portion **233** opening **238** reservoir space **239** scale mark **300** liquid collector **310** flow passage member **311** outside tube **312** inside member **313** flow passage **320** contact member **321** flow-passage-member receiving hole **323** contact surface **330** container **331** peripheral wall **332** bottom portion **333** opening **338** reservoir space **339** scale mark **340** inner member **341** cone-shaped portion **400** liquid collector **410** flow passage member **411** outside tube **412** inside member **413** flow passage **420** contact member **421** flow-passage-member receiving hole **423** contact surface **425** sleeve **430** container **431** peripheral wall **432** bottom portion **433** opening **438** reservoir space **439** scale mark **440** inner member **441** cone-shaped portion **500** liquid collector **510** flow passage member **511** outside tube **512** inside member **513** flow passage **530** container **531** peripheral wall **532** bottom portion **533** opening **538** reservoir space **539** scale mark **540** inner member **541** cone-shaped portion **610** flow passage member **613** flow passage **615** groove **710** flow passage member **713** flow passage **715** through hole **715A** through hole **715B** through hole **810** flow passage member **812** single pipe **812A** single pipe **812B** single pipe **813** flow passage **815** hole **816** gap **880** flow passage member **882** pipe **883** flow passage **885** hole **910** flow passage member **911** outer tube **912** inner pipe **912A** inner pipe **912B** inner pipe **913** flow passage **915** hole **916** gap **917** outer peripheral gap **999** liquid **1000** surface **1101** one end **1102** other end **1111** inner peripheral surface **1121** groove **1122** outer peripheral surface **1211** inlet opening **1212** outlet opening **2101** one end **2102** other end **2111** inner peripheral surface **2121** groove **2122** outer peripheral surface **2211** inlet

opening 2212 outlet opening 2314 inner peripheral surface 3101 one end 3102 other end 3111 inner peripheral surface 3121 groove 3122 outer peripheral surface 3211 inlet opening 3212 outlet opening 3411 first opening 3412 second opening 4101 one end 4102 other end 4111 inner peripheral surface 4121 groove 4122 outer peripheral surface 4211 inlet opening 4212 outlet opening 4411 first opening 4412 second opening 5101 one end 5102 other end 5111 inner peripheral surface 5121 groove 5122 outer peripheral surface 5411 first opening 5412 second opening 6122 outer peripheral surface 9111 inner peripheral surface 9122 outer peripheral surface
Ds pipe axis direction Dv vertical direction

Claims

1. A liquid collector for collecting liquid, the liquid collector comprising: a flow channel member including a flow channel that connects one end and other end of the flow channel member; a contact member disposed at the one end of the flow channel member, and that includes an elastic material; and the contact member comprising: an inlet opening configured to come into contact with a surface of the target from which the liquid is to be collected and to allow introduction of the liquid that is on or near the surface, and an outlet opening connected to the inlet opening to provide fluid communication and also connected to the flow channel member to provide fluid communication, wherein the one end of the flow channel member is fitted to the outlet opening of the contact member; a container configured to receive the liquid discharged from the flow channel at the other end of the flow channel member; and a cone-shaped portion disposed in the container, wherein at least a portion of the cone-shaped portion in the pipe axis direction has an inner diameter that decreases toward a bottom portion in the pipe axis direction, wherein the other end of the flow channel member is in contact with an inner peripheral surface of the cone-shaped portion.
2. The liquid collector according to claim 1, wherein the contact member includes a sleeve that has a tubular shape and extends toward the other end of the flow channel member.
3. The liquid collector according to claim 1, wherein the contact member has a contact surface in a region around the inlet opening, the contact surface coming into contact with the surface of the target from which the liquid is to be collected.
4. The liquid collector according to claim 3, wherein at least a portion of the contact surface is spherical.
5. The liquid collector according to claim 1, wherein the flow channel member includes a plurality of the flow channels that extend in a pipe axis direction connecting the one end and the other end of the flow channel member.
6. The liquid collector according to claim 5, wherein the plurality of the flow channels are substantially parallel to each other.
7. The liquid collector according to claim 5, wherein the flow channel member has a columnar shape that extends in the pipe axis direction connecting the one end and the other end of the flow channel member, and has a plurality of grooves that extend in the pipe axis direction along an outer peripheral surface of the flow channel member.
8. The liquid collector according to claim 5, wherein the flow channel member includes an outside tube having a tubular shape that extends in the pipe axis direction, and an inside member inserted in the outside tube and having a plurality of grooves that extend in the pipe axis direction along an outer peripheral surface of the flow channel member, and wherein an inner peripheral surface of the outside tube and the plurality of grooves in the inside member define the plurality of the flow channels.
9. The liquid collector according to claim 5, wherein the flow channel member has a columnar shape that extends in the pipe axis direction, and the plurality of the flow channels extend through the flow channel member in the pipe axis direction to provide communication between the one end and the other end of the flow channel member.

10. The liquid collector according to claim 5, wherein the flow channel member includes a plurality of single pipes that are bundled together and that extend in the pipe axis direction.
 11. The liquid collector according to claim 5, wherein the flow channel member includes: an outer tube having a tubular shape that extends in the pipe axis direction, and a plurality of inner pipes that are inserted in the outer tube and that extend in the pipe axis direction.
 12. The liquid collector according to claim 5, wherein the container has an opening at a side opposite to a side at which the bottom portion is provided in the pipe axis direction, and wherein the contact member is disposed to close the opening.
 13. The liquid collector according to claim 1, wherein the container includes: a peripheral wall having a tubular shape that extends in the pipe axis direction, and a bottom portion that blocks the peripheral wall at one side in the pipe axis direction.
 14. The liquid collector according claim 13, wherein the bottom portion of the container is spaced from the other end of the flow channel member in the pipe axis direction, and wherein a reservoir space to store the liquid is defined between the bottom portion of the container and the other end of the flow channel member.
 15. The liquid collector according claim 13, wherein the peripheral wall of the container has an inner diameter that gradually decreases toward the bottom portion in the pipe axis direction.
 16. The liquid collector according to claim 13, wherein the other end of the flow channel member is in contact with an inner peripheral surface of the peripheral wall of the container.
 17. The liquid collector according to claim 13, wherein the container has an air hole that provides communication with an outside of the container.
 18. The liquid collector according to claim 13, wherein the container has a scale mark to indicate an amount of the liquid received in the container.
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